

Band picture: conductor vs semiconductor vs insulator

In solids, discrete atomic levels broaden into **bands**. The two most relevant are: **valence band (VB)** and **conduction band (CB)**. Their separation is the **bandgap E_g** . (Details will be in EE2103 Semiconductor Device Fundamentals)

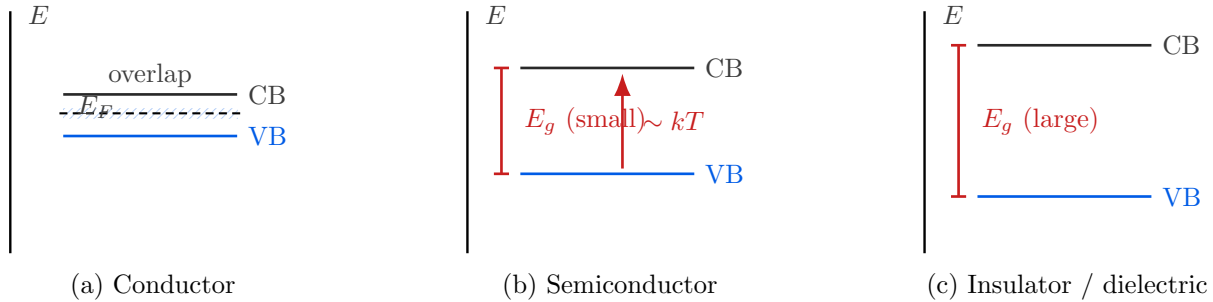


Figure 1: **Band-based classification.** (a) Conductors have band overlap / partially filled states $\Rightarrow$ many free carriers. (b) Semiconductors have a small E_g so some carriers can be generated (thermal $\sim kT$ or via doping). (c) Insulators (dielectrics) have a large E_g so VB stays full and CB empty at room temperature.

Dielectric response: induced dipole (no free electrons)

For a **dielectric**, electrons remain *bound* to atoms/molecules (they do not move freely across the crystal as in conductors). When an electric field $\vec{E}$ is applied, orbitals distort slightly and the electron cloud shifts relative to the nucleus (The charges are *not free* to leave the material, but a small separation occurs): an **induced dipole** forms.

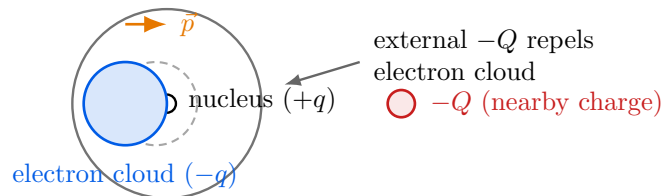


Figure 2: A nearby negative charge polarizes the atom: the electron cloud shifts away, leaving the positive side closer to $-Q$, forming an induced dipole (p : dipole moment).

Applying Gauss's law to the induced field and defining $\mathbf{D}$

Coulomb/Gauss law always uses *total* charge. In electrostatics, the electric field at any point is produced by the *total* charge distribution:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \iiint \rho_{\text{total}}(\mathbf{r}') \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3} d^3r', \quad \rho_{\text{total}} = \rho_f + \rho_b,$$



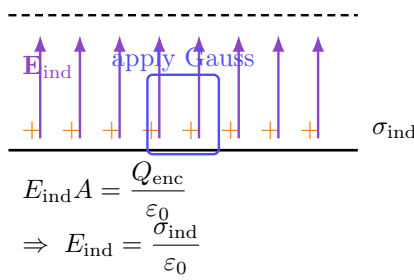
where ρ_f is **free charge density** and ρ_b is **induced/bound charge density**. Equivalently (differential Gauss law),

$$\nabla \cdot \mathbf{E} = \frac{\rho_f + \rho_b}{\varepsilon_0} \quad (1)$$

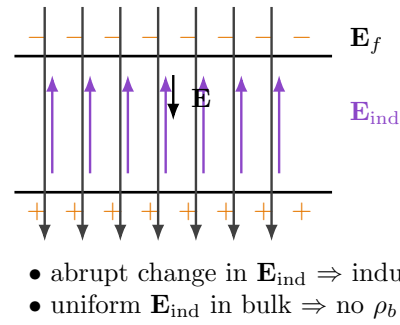
Conceptual split. We write the total field as

$$\mathbf{E} = \mathbf{E}_f + \mathbf{E}_{\text{ind}},$$

where $\mathbf{E}_f$ is the field due to *free charges* and $\mathbf{E}_{\text{ind}}$ is the field due to *induced/bound charges* in the dielectric (opposite direction).



(a) Pillbox across induced surface charge σ_{ind} .



(b) Black: $\mathbf{E}_f$, Blue: $\mathbf{E}_{\text{ind}}$, Net: $\mathbf{E} = \mathbf{E}_f + \mathbf{E}_{\text{ind}}$.

Figure 3: Use Gauss's law on $\mathbf{E}_{\text{ind}}$ to relate it to induced charge, then use divergence relations to motivate $\mathbf{D}$.

Induced charge appears where $\mathbf{E}_{\text{ind}}$ has divergence. By definition,

$$\nabla \cdot \mathbf{E}_{\text{ind}} = \frac{\rho_b}{\varepsilon_0} \quad (2)$$

(If $\mathbf{E}_{\text{ind}}$ is uniform in a region, then $\nabla \cdot \mathbf{E}_{\text{ind}} = 0$ there, i.e., no bound *volume* charge in that region.)

Subtract to isolate free charge but using $\mathbf{E}$. From (1) and (2):

$$\nabla \cdot \mathbf{E} - \nabla \cdot \mathbf{E}_{\text{ind}} = \frac{\rho_f}{\varepsilon_0} \Rightarrow \nabla \cdot (\varepsilon_0 (\mathbf{E} - \mathbf{E}_{\text{ind}})) = \rho_f \quad (3)$$

Now define the **displacement field**

$$\mathbf{D} \equiv \varepsilon_0 (\mathbf{E} - \mathbf{E}_{\text{ind}}) \Rightarrow \nabla \cdot \mathbf{D} = \rho_f$$

This is the key “trick”: $\mathbf{D}$ is built so that Gauss's law involves only **free charge**.

Linear isotropic dielectric Experiments show (in the linear regime) that the induced field is proportional to $\mathbf{E}$ and opposite to it:

$$\mathbf{E}_{\text{ind}} = -\chi_e \mathbf{E}$$

Substitute into $\mathbf{D} = \varepsilon_0(\mathbf{E} - \mathbf{E}_{\text{ind}})$:

$$\mathbf{D} = \varepsilon_0(1 + \chi_e)\mathbf{E} = \varepsilon_0\varepsilon_r \mathbf{E}, \quad \varepsilon_r = 1 + \chi_e$$

(Manufacturers provide ε_r for the material in the specified linear field range.)

Takeaway. $\nabla \cdot \mathbf{E} = (\rho_f + \rho_b)/\varepsilon_0$ uses *total* charge. Defining $\mathbf{D} \equiv \varepsilon_0(\mathbf{E} - \mathbf{E}_{\text{ind}})$ gives $\nabla \cdot \mathbf{D} = \rho_f$. For linear isotropic dielectrics: $\mathbf{E}_{\text{ind}} = -\chi_e \mathbf{E}$ and $\mathbf{D} = \varepsilon_0(1 + \chi_e)\mathbf{E} = \varepsilon_0\varepsilon_r \mathbf{E}$.

Why we assume linear & isotropic response

(1) Microscopic picture & why d is “unknown” analytically. From the pillbox argument we obtained

$$E_{\text{ind}} = \frac{\sigma_{\text{ind}}}{\varepsilon_0}.$$

Microscopically, polarization arises because the electron cloud shifts slightly relative to the nucleus by an average distance d . That shift produces a dipole moment $p = qd$ for each atom/molecule, and collectively leads to an induced surface charge density σ_{ind} . However, the exact value of d depends on the atomic binding potential and is not convenient to compute from first principles for most materials. So, we use the experimentally observed relationship between d and the applied field.

(2) Linear (engineering) regime. Experiments show that as the field increases, the electron cloud shift increases; for small/moderate fields, the response is approximately linear:

$$d \propto E \Rightarrow \sigma_{\text{ind}} \propto d \propto E \Rightarrow E_{\text{ind}} = \frac{\sigma_{\text{ind}}}{\varepsilon_0} \propto E.$$

Hence, in most engineering applications (within the linear range),

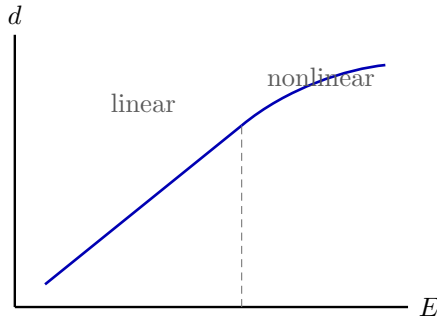
$$E_{\text{ind}} \propto E \quad (\text{linear dielectric assumption}).$$

(3) Precautionary note: anisotropic materials. In some materials, the induced response need not align with the direction of the applied field. That is, the electron-cloud shift (and hence induced field/polarization) may not be along $\vec{E}$:

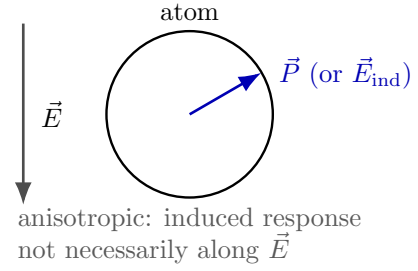
$$\vec{P} \nparallel \vec{E} \quad (\text{anisotropic medium}).$$

In such cases one needs a tensor relation (conceptually): $\vec{P} = \varepsilon_0 \chi \vec{E}$. But here we assume an isotropic dielectric, so the induced response is along the field direction:

$$\vec{P} \parallel \vec{E}, \quad \vec{E}_{\text{ind}} \parallel \vec{E} \quad (\text{isotropic assumption}).$$



(a) Typical trend: d vs E (linear range used).



(b) Precaution: anisotropic response ($\vec{P} \nparallel \vec{E}$).

Take away

- $\vec{E}$: applied electric field.
- $\vec{E}_{\text{ind}}$: induced field due to bound surface charges (opposes $\vec{E}$).
- σ_{ind}^- , σ_{ind}^+ : induced *surface charge densities* (units: C/m²), appearing at the top and bottom faces.
- If n is the number density (units: number/m³) and the microscopic displacement is d , then the surface bound charge density scales as

$$\sigma_{\text{ind}}^- \sim (-e) n d, \quad \sigma_{\text{ind}}^+ \sim (+e) n d.$$

(Here e is the elementary charge.)

- Dielectrics do not have free-charge drift in electrostatics; an applied field mainly *distorts* bound charges.
- Each atom/molecule develops an induced dipole (small separation d).
- In a polarized block, interior dipoles cancel, leaving only bound surface charge, which creates $\vec{E}_{\text{ind}}$ opposing $\vec{E}$.

Problem setup

A point charge Q is placed at the center of a spherically layered medium:

Region I: $0 < r < a$ (air) Region II: $a < r < b$ (glass) Region III: $r > b$ (air).

We assume a linear isotropic dielectric in Region II with relative permittivity ϵ_r (for glass, typically $\epsilon_r \approx 2.25$).



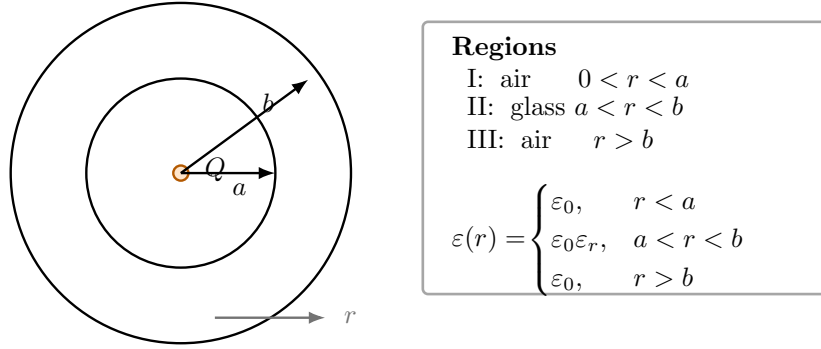


Figure 4: Spherical layered medium with central free charge Q : air cavity ($r < a$), glass shell ($a < r < b$), and air outside ($r > b$).

Step 1: Use Gauss's law in terms of $\vec{D}$ (this is the key “usefulness”)

Even though the induced/bound charges in the glass make $\vec{E}$ less obvious in Region II, the displacement field satisfies Gauss's law involving **only free charge**:

$$\nabla \cdot \vec{D} = \rho_f.$$

By spherical symmetry, $\vec{D} = D_r(r) \hat{r}$. For any Gaussian sphere of radius $r > 0$,

$$\oint \vec{D} \cdot d\vec{S} = D_r(r) 4\pi r^2 = Q \quad \Rightarrow \quad \boxed{D_r(r) = \frac{Q}{4\pi r^2}} \quad (\text{valid for all } r > 0).$$

This is exactly why $\vec{D}$ is useful: D_r is **obtained immediately** without solving for induced charges.

Step 2: Convert $\vec{D}$ to $\vec{E}$ in each region

For a linear isotropic medium, $\vec{D} = \varepsilon \vec{E}$, hence

$$\vec{E}(r) = \frac{\vec{D}(r)}{\varepsilon(r)}.$$

Therefore,

$$\vec{E}(r) = \begin{cases} \frac{Q}{4\pi\varepsilon_0 r^2} \hat{r}, & 0 < r < a \quad (\text{air}) \\ \frac{Q}{4\pi\varepsilon_0 \varepsilon_r r^2} \hat{r}, & a < r < b \quad (\text{glass}) \\ \frac{Q}{4\pi\varepsilon_0 r^2} \hat{r}, & r > b \quad (\text{air}) \end{cases}$$

So the field magnitude is reduced by a factor ε_r inside the glass shell.

Step 3: Potential everywhere (with $V(\infty) = 0$)

Since $E_r = -dV/dr$ and the field is radial,

$$V(r) = - \int_{\infty}^r E_r(r') dr'.$$

Compute piecewise:

Region III: $r > b$ (air)

$$V(r) = \frac{Q}{4\pi\epsilon_0 r}, \quad r > b.$$

Region II: $a < r < b$ (glass)

Use continuity of potential and integrate from b to r :

$$V(r) = V(b) + \int_r^b \frac{Q}{4\pi\epsilon_0\epsilon_r r'^2} dr' = \frac{Q}{4\pi\epsilon_0 b} + \frac{Q}{4\pi\epsilon_0\epsilon_r} \left(\frac{1}{r} - \frac{1}{b} \right).$$

$$V(r) = \frac{Q}{4\pi\epsilon_0 b} + \frac{Q}{4\pi\epsilon_0\epsilon_r} \left(\frac{1}{r} - \frac{1}{b} \right), \quad a < r < b.$$

Region I: $0 < r < a$ (air cavity)

Again use continuity at $r = a$:

$$V(r) = V(a) + \int_r^a \frac{Q}{4\pi\epsilon_0 r'^2} dr' = V(a) + \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{1}{a} \right),$$

where

$$V(a) = \frac{Q}{4\pi\epsilon_0 b} + \frac{Q}{4\pi\epsilon_0\epsilon_r} \left(\frac{1}{a} - \frac{1}{b} \right).$$

So,

$$V(r) = \frac{Q}{4\pi\epsilon_0 b} + \frac{Q}{4\pi\epsilon_0\epsilon_r} \left(\frac{1}{a} - \frac{1}{b} \right) + \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{1}{a} \right), \quad 0 < r < a.$$

$$D_r = \frac{Q}{4\pi r^2} \text{ everywhere,} \quad \epsilon_0 E_r = \begin{cases} D_r, & r < a \text{ and } r > b \\ D_r/\epsilon_r, & a < r < b \end{cases}$$

so $\epsilon_0 E_r$ drops inside the glass region.



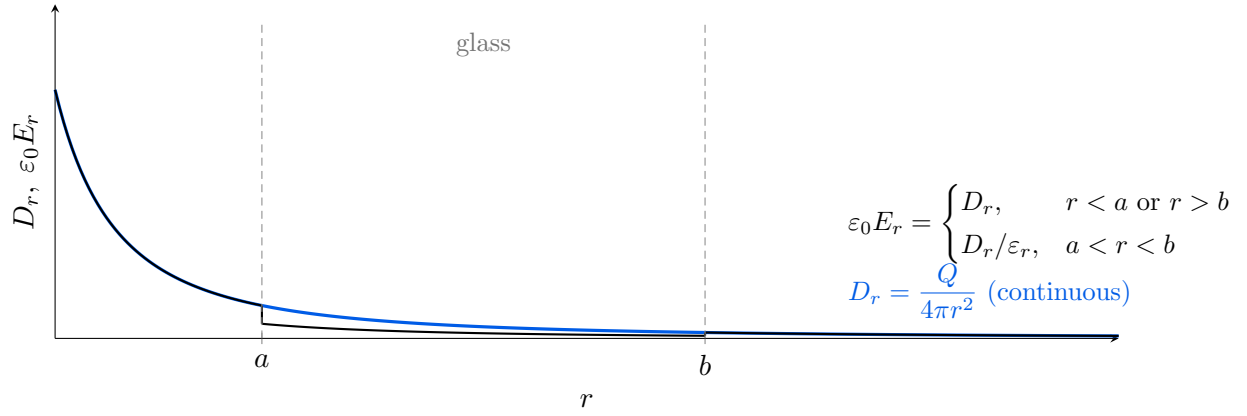


Figure 5: Neat conceptual plot (normalized with $Q/(4\pi) = 1$). D_r is continuous across a and b , while E_r is reduced inside glass by ε_r and hence shows jumps at the interfaces.

Key takeaway

Even when $\vec{E}$ inside a dielectric is not obvious due to induced charges, $\vec{D}$ is straightforward from free charge:

$$\oint \vec{D} \cdot d\vec{S} = Q_{\text{free, enclosed}} \Rightarrow \vec{E} = \vec{D} / \varepsilon \text{ (region-wise).}$$